

WEEK 9

Week 9A — Low-Level File Copy and Performance

This program copies a file using the low-level system calls `open()`, `read()`, `write()`, `lseek()`, and `close()`.

```
GNU nano 8.7.1 week9a.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <time.h>

int main()
{
    int src, dest;
    char buffer[1024];
    int bytes;
    clock_t start, end;

    src = open("input.txt", O_RDONLY);

    if (src < 0)
    {
        printf("Error opening input file\n");
        return 1;
    }

    dest = open("output.txt",
               O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (dest < 0)
    {
        printf("Error opening output file\n");
        close(src);
        return 1;
    }

    start = clock();

    while ((bytes = read(src, buffer, sizeof(buffer))) > 0)
    {
        write(dest, buffer, bytes);
    }
}
```

```
sri_dhruthi@Dhruthi:~$ nano week9a.c
sri_dhruthi@Dhruthi:~$ nano input.txt
sri_dhruthi@Dhruthi:~$ gcc week9a.c -o week9a
sri_dhruthi@Dhruthi:~$ ./week9a
File copied successfully.
Time taken: 0.000056 seconds
sri_dhruthi@Dhruthi:~$ cat output.txt
Operating System Lab
File Copy using System Calls
```

Week 9B — I/O Redirection Using `dup2()`

This program redirects standard input/output using file descriptors and `dup2()`. It demonstrates how a program can read from or write to a file instead of the terminal.

```
sri_dhruthi@Dhruthi: ~  
GNU nano 8.7.1 week9b.c *  
#include <stdio.h>  
#include <fcntl.h>  
#include <unistd.h>  
  
int main()  
{  
    int fd;  
  
    fd = open("redirect.txt",  
             O_WRONLY | O_CREAT | O_TRUNC, 0644);  
  
    if (fd < 0)  
    {  
        printf("File opening failed\n");  
        return 1;  
    }  
  
    dup2(fd, STDOUT_FILENO);  
  
    close(fd);  
  
    printf("This output is redirected to the file.\n");  
    printf("dup2() redirects standard output.\n");  
  
    return 0;  
}
```

```
sri_dhruthi@Dhruthi:~$ nano week9b.c  
sri_dhruthi@Dhruthi:~$ gcc week9b.c -o week9b  
sri_dhruthi@Dhruthi:~$ ./week9b  
sri_dhruthi@Dhruthi:~$ cat redirect.txt  
This output is redirected to the file.  
dup2() redirects standard output.  
sri_dhruthi@Dhruthi:~$ |
```